

Question		Answer	Mark	Mark scheme	Additional guidance
17	(a)(i)	$(x - 4)^2 - 13$	M1	for start to method of completing the square, eg $(x - 4)^2 \dots$	This mark is dependent on a completed square form given in (a)(i) Condone one sign error One correct solution in the required form out of two solutions given scores 2 marks. If an answer is shown in the range in working and then incorrectly rounded award full marks.
	(ii)	$(4, -13)$	A1 B1	cao ft from answer of form $(x - a)^2 - b$	
	(b)	0.449 and -1.59	M1	for substitution into quadratic formula, eg $\frac{-8 \pm \sqrt{8^2 - 4 \times 7 \times -5}}{2 \times 7}$ oe	
			M1	for simplification to $\frac{-8 \pm \sqrt{204}}{14}$ or $\frac{-4 \pm \sqrt{51}}{7}$ oe or for one correct solution in decimal form.	
			A1	for 0.448 to 0.449 and -1.59 to $-1.6(0)$	
	(c)	Explanation	C1	for explanation Acceptable He hasn't considered the coefficient of 3 He forgot about the 3 The answer $x = 2$ should be $x = \frac{2}{3}$ The first bracket was not $k - 2$ $k = 2$ is wrong, he should have divided by 3 Not acceptable One of the answers is wrong The solutions are not correct He should have written $3(k - 2)(k + 4)$ The $3k$ should not be inside the bracket 2 added to -4 does not give 10 $k = 2$ is wrong	